Project Synopsis Report (Phase I – 5th Semester)

of the

Final Year Project

titled

"Design and Simulation of a Hybrid CMOS–SET Based Operational Amplifier"

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1. Abstract

CMOS scaling is approaching fundamental physical limits, motivating research into post-CMOS devices such as Single-Electron Transistors (SETs). SETs offer extremely low power dissipation via the Coulomb blockade effect, but suffer from poor current drivability, low gain, and high output impedance, limiting standalone use. Hybrid CMOS–SET circuits combine SET-level power efficiency with CMOS drive strength and have proven successful in digital logic, though prior designs modelled the SET and CMOS devices separately — using the MIB compact model and BSIM4.6.1 respectively — within a SPICE-based (T-Spice) environment. This project extends the hybrid approach to the analog domain, proposing a hybrid CMOS–SET Operational Amplifier that will instead be implemented and simulated as a single, unified model within MATLAB/Simulink, with its gain, bandwidth and power dissipation benchmarked against a conventional CMOS op-amp.

2. Introduction

A) Overview

Moore's Law is nearing physical limits as MOSFET channel lengths shrink to a few nanometres, driving interest in post-CMOS technologies. Single-Electron Transistors (SETs) — three-terminal devices where a gate-controlled island exchanges electrons with source and drain through tunnel junctions (Fig. 1) — promise atomic-scale density and ultra-low power via the Coulomb blockade effect. However, a standalone SET has poor current drivability and very high output impedance, so it cannot directly replace MOSFETs. Pairing an SET with a CMOS transistor as load/driver produces a hybrid CMOS–SET circuit that keeps the SET's low-power switching while using CMOS for output drive. This project extends that hybrid concept — so far mainly demonstrated for digital logic — to an analog Operational Amplifier.

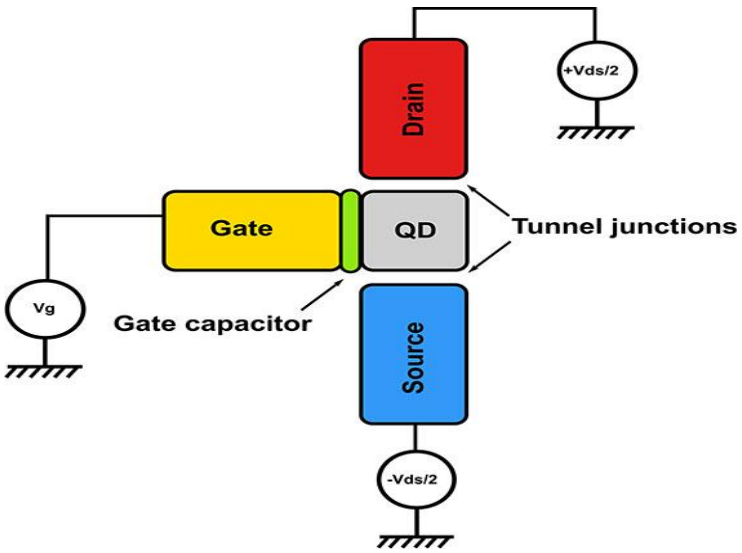


Fig. 1: Structure of a Single-Electron Transistor (SET) — source, drain and gate coupled to a conducting island via tunnel junctions.

B) Prior Work (Literature Review)

Jana et al. (2013) demonstrated a room-temperature-operable hybrid CMOS–SET inverter using a PMOS load with an SET pull-down device, verified with the MIB compact SET model and the BSIM4.6.1 CMOS model in a Tanner/T-Spice environment [1]. Extending this inverter, they realised NAND, NOR, AND, OR, XOR and XNOR gates using series–AND / parallel–OR connection rules, all

functioning correctly at a room-temperature-operable 800 mV supply. Notably, the hybrid inverter achieved a voltage gain of about 4.32–9.20 (for W/L = 1.5–10) and consumed roughly one-sixth the power of a conventional CMOS inverter (Fig. 2) [1]. These results confirm hybridisation's feasibility for digital logic; extending the same principle to analog blocks such as op-amps remains largely unexplored, motivating this project.

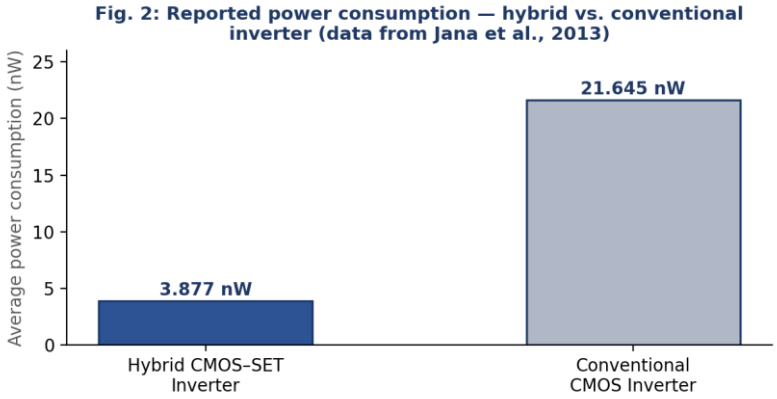


Fig. 2: Reported average power consumption, hybrid CMOS–SET vs. conventional CMOS inverter (data from [1]).

C) Challenges

Key challenges: SETs exhibit non-linear, staircase-like I–V behaviour unsuited to smooth analog gain, unlike their clean switching behaviour in logic gates; high sensitivity to background charge demands carefully chosen room-temperature-operable parameters; MATLAB/Simulink has no built-in SET model, so one must be built from published MIB equations; and fair benchmarking against a conventional CMOS op-amp requires consistent bias and sizing conditions.

D) Motivation

Battery-operated and energy-harvested electronics — wearables, implantable sensors, IoT nodes — demand analog blocks that consume far less power than conventional CMOS. Hybrid CMOS–SET logic has already shown roughly six-fold power savings over CMOS at room temperature; achieving a similar gain for an op-amp — one of the most fundamental analog building blocks — would be a meaningful step towards ultra-low-power mixed-signal design, an area so far dominated by digital-logic studies.

E) Objective

- Study SET operation, the Coulomb blockade, and the MIB compact model.
- Review hybrid CMOS–SET logic topologies for design principles transferable to analog circuits.
- Propose a room-temperature-operable hybrid CMOS–SET op-amp topology.
- Model and simulate the design in MATLAB/Simulink.
- Characterise DC transfer, open-loop gain, bandwidth and power dissipation.
- Benchmark performance against a conventional CMOS op-amp.

3. Requirement Analysis

A) Functional Requirement

- Shall amplify the input voltage linearly within its operating range.
- Shall function correctly at room temperature using literature-consistent SET parameters.
- Shall interface at CMOS-compatible voltage levels (e.g., ~0.8 V, scaled as needed).
- Shall maintain a stable DC bias point prior to AC/transient analysis.

B) System Requirement

- An SET-based active load/gain element combined with CMOS stages for biasing and output drive (Fig. 3).
- Room-temperature-operable SET and CMOS parameter sets adapted from literature.
- A simulation environment able to solve the combined SET–CMOS circuit equations.

Fig. 3: Proposed block-level architecture of the hybrid CMOS–SET op-amp

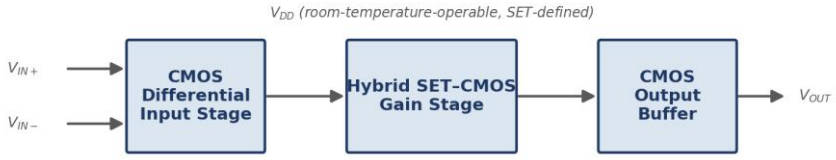


Fig. 3: Proposed block-level architecture of the hybrid CMOS–SET op-amp.

C) Hardware Requirement

- A standard computer (8 GB+ RAM) running MATLAB/Simulink.

D) Software Requirement

- MATLAB/Simulink; Simscape Electrical or custom S-function blocks for SET behaviour (no built-in SET block exists); standard documentation tools.

4. Feasibility Analysis

A) Technical Feasibility

Prior literature confirms hybrid CMOS–SET circuits can be modelled and simulated successfully at room temperature (Section 2B), and the team's existing HDL, FPGA and simulation experience supports building and debugging the MATLAB/Simulink model.

B) Scheduling Feasibility

The project follows four broad stages — literature review (Phase I), topology/model development, simulation and characterisation, and documentation — aligned with the Phase I/II/III evaluation structure, allowing time for SET macro-model debugging.

C) Financial Feasibility

As a simulation-only project with no fabrication, the only requirement is MATLAB/Simulink access, already available through the institute — financially negligible.

D) Social and Ethical Consideration

No human/animal subjects or environmental impact are involved. A successful low-power hybrid op-amp would support the broader goal of energy-efficient electronics; all prior work is properly cited.

5. Methodology

1. Survey SET physics, the MIB model, and hybrid CMOS–SET logic designs.
2. Select a standard CMOS op-amp topology as baseline.
3. Define a hybrid SET–CMOS topology and room-temperature-operable parameters.
4. Build the model in MATLAB/Simulink (Simscape + custom SET macro-model).
5. Run DC, AC and transient simulations to extract gain, bandwidth and power.
6. Compare results against the conventional CMOS baseline and document findings.

Fig. 4: Proposed methodology workflow

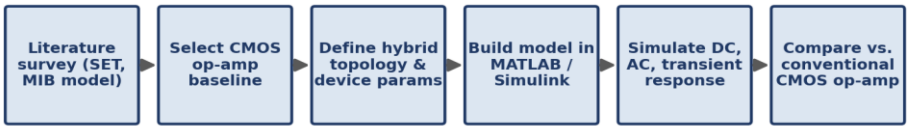


Fig. 4: Proposed methodology workflow.

6. Future Scope

1. Extend the design to a fully differential, multi-stage topology for improved CMRR and gain.
2. Study sensitivity to background charge and temperature beyond the room-temperature-operable regime considered here.
3. Explore layout and fabrication feasibility using room-temperature-operable SOI-based SETs.
4. Integrate the validated op-amp into larger low-power blocks (ADCs, filters, sensor interfaces) for IoT applications.
5. Benchmark the hybrid approach against other emerging post-CMOS candidates such as FinFET and TFET.

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